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ANTI HISTAMINIC

AGENTS

Syllabus →

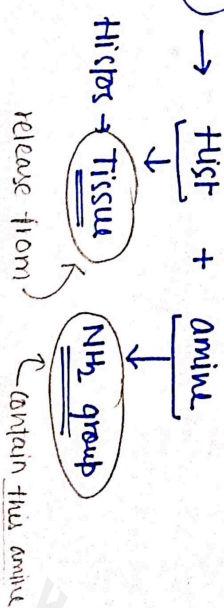
- Antihistaminic agents: histamine, receptors and their distribution in the human body.
- H_1 - Antagonists
- H_2 - Antagonists
- Gastric proton pump inhibitors
- Antihistaminic agents ⇒ These are those agents or drugs which inhibit the action of histamine in human body.

• Histamine → It is a chemical substance present in our body and it comes under the class of Autacoids



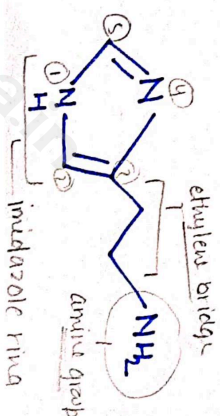
• Autacoids → these are the local hormone, which release from the tissue and act at the site of synthesis and release.

• Histamine → Hist + amine



• Histamine →

2 - (1H-Imidazol-4-yl)-ethanamine



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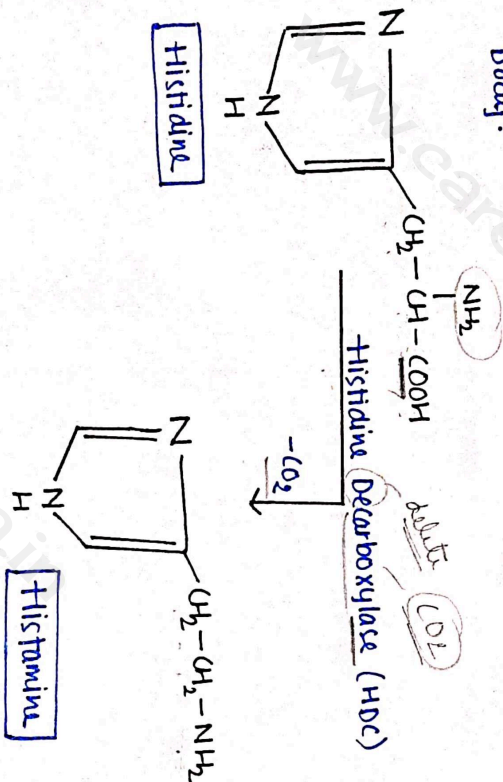
- Histamine is discovered by Sir Henry

Dale in 1910.

— Biosynthesis, Storage, Release and Catabolism
of histamine

Synthesis →

Histamine is synthesized by decarboxylation of histidine which is found in all organs and tissue of the human body.



Storage →

Histamine mostly present inside the

mast cells and basophils

- Both mast cells and basophils are the type of white blood cells and a part of immune system.

- It is also stored in enterochromaffin-like (ECL) cell and a variety of neurons.

① ECL

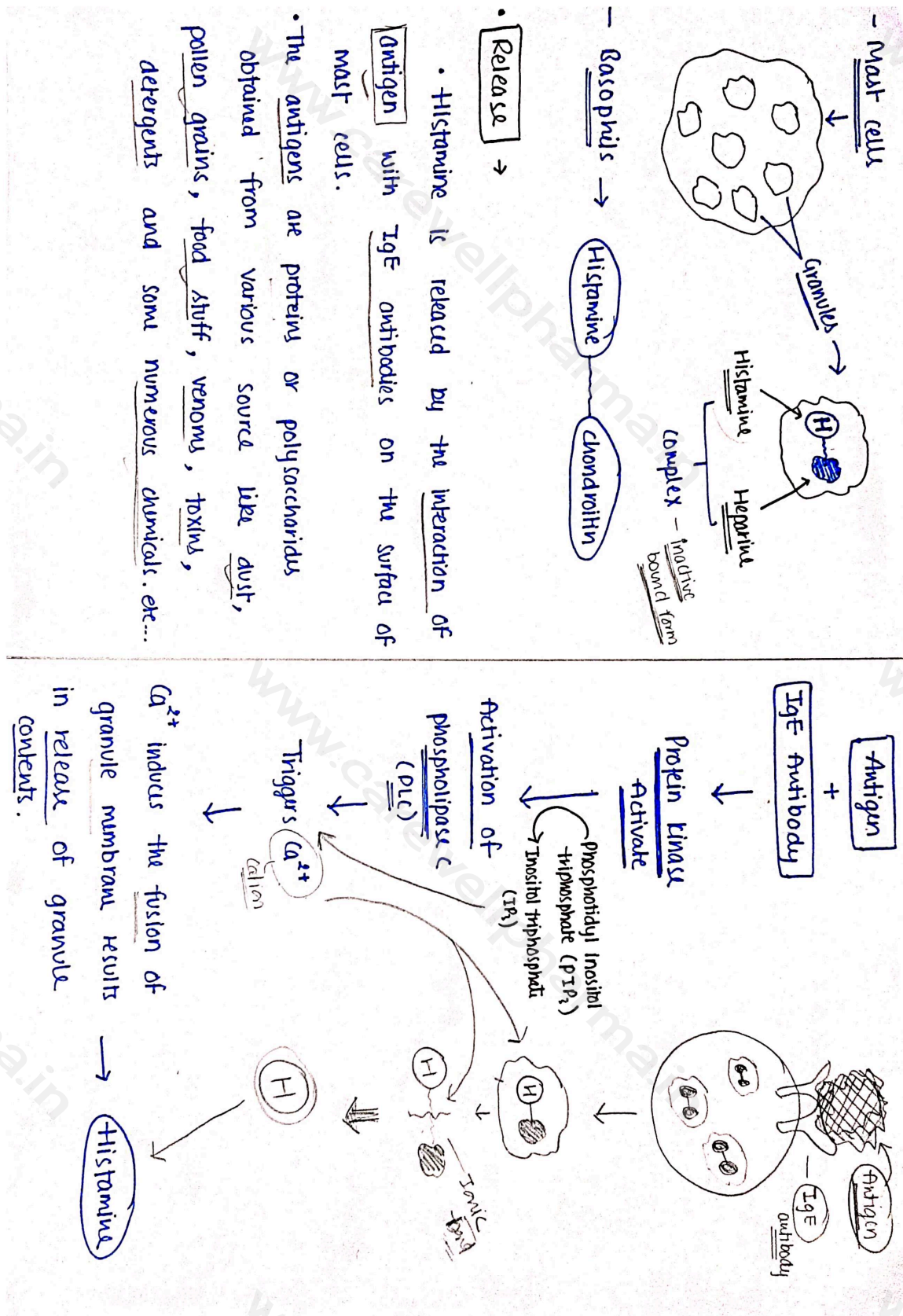
② histaminergic neurons

- Usually histamine (mast cells) found in throughout the body, but in high concentration is present in skin, mucosal cell of the lungs, intestine, urinary tract and tissue adjacent to the circulation.

- Its concentration is also found high in mammalian Cerebro Spinal fluid (brain).

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— there are also some other inflammatory mediators release such as prostaglandins, leukotrienes, kinins and platelet activating factor.

⊙ After release of histamine from mast cells it performs following physiological effects :-

- Hyper sensitivity
 - itching, sneezing, watery eye and running nose.
- Contracts smooth muscles of the lungs.
- tissue injury ✓
- inflammation ✓ — cause pain
- Allergic reactions ✓
- Some physiological role of histamine →
 - Bronchospasm — cause due to irritation,
 ↓
inflammation, or allergic reaction
 ↓
hard to breathe of the airways.

- Vasodilation → cause hypotension.
- Gastric secretion → increase HCl secretion
 eg. acidity ↑, ulcer ↑ etc...
- produce oedema → fluid and plasma protein may come to extracellular fluid.
- etc---
- Catabolism →
 - Histamine inactivate by many reactions and enzyme which change the structure of histamine.
 - After changing structure, histamine does not bind with receptor due to change its structure.

[eg] Histamine $\xrightarrow[\text{oxidative deamination}]{\text{Imidazole-N-methyl transferase}}$ Methyl histamine $\xrightarrow[\text{MAO}]{\text{Methyl histamine}}$ Methyl Imidazole $\xrightarrow{\text{fretic acid}}$ fretic acid

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HISTAMINE RECEPTORS

There are mainly four receptors, in which histamine bind and give their pharmacological effects :-



All these receptors are G-protein coupled receptor and give their action by the mechanism of GPCR.

• Distributions →

① $H_1 \Rightarrow$ stimulates phospholipase C, - PLC

• Location :-

- smooth muscle of respiratory and digestive GI tract Lungs
- CNS neurons
- vascular endothelial cells, in heart
- T-cells, B-cells, neutrophils, eosinophils

• Effects →

- Pain, Pruritus (Itching), Vasodilation, hypotension, headache, Bronchoconstriction
- Increased vascular permeability at inflammation sites
- Triple response
- oedema formation
- severe allergic response
- IgE production.

② $H_2 \Rightarrow$ stimulates adenylylase and increase cAMP.

• Location :-

- Gastric parietal cells
- vascular smooth muscles
- CNS, heart and uterus.

• Effects :-

- increase in gastric acid secretion (HCl).
- vascular permeability, vasodilation.

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③ $H_3 \Rightarrow$

• Location :-

- mostly found in CNS.

• Effects :-

- inhibition of histamine release

(pre synaptic auto receptor)

- modulates the release of 5-HT, dopamine, NAD, Ach and GABA in CNS by acting as heteroreceptor.

④ $H_4 \Rightarrow$

• Location :-

- Hematopoietic cells

↳ the stem cells that give rise to other blood cells.

• Effects :-

- modulate immune function.

CLASSIFICATION OF ANTIHISTAMINES

H_1 - Antagonists

↓

• First Generation

- Diphenhydramine HCl *
- Dimenhydrinate
- Doxylamine succinate
- Clemastine fumarate
- Diphenyl pyridine HCl
- Tripeleamine HCl
- Chlorcyclizine HCl
- Meclizine HCl
- Budizine HCl
- Chlorpheniramine maleate
- Triprolidine HCl *
- Pheniramine tartrate
- Promethazine HCl *

H_2 - Antagonists

↓

• Gastric Acid Inhibitors

- cimetidine *
- famotidine
- Ranitidine ✓

• Gastric Proton Pump inhibitors (PPI)

- Omeprazole -
- Lansoprazole
- Rabeprazole
- Pantoprazole

- Trimoprazole tartrate
- Cyproheptadine HCl
- Azatidine maleate

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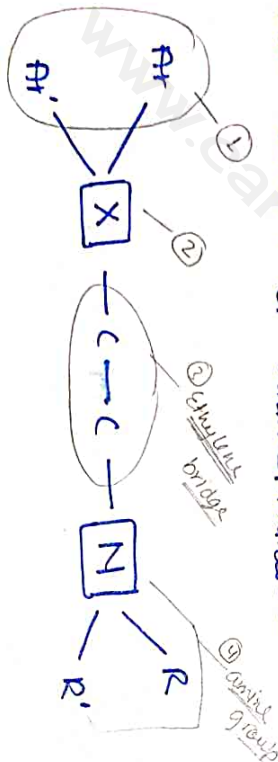
Second generation

- Astemizole
- Loratadine
- Cetirizine
- Levocetirizine
- Mast cell stabilizer
- Cromolyn sodium

SAR OF ANTI-HISTAMINES

⇒ for H_1 -Antagonists

Basic structure of Antihistamines -



- Structurally, Substitution (modification) is possible on :-

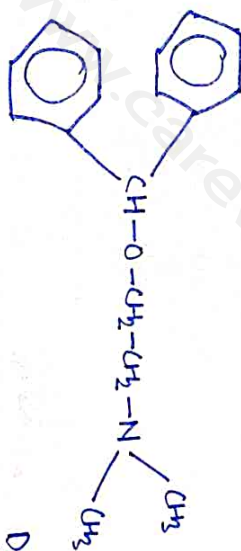
- Aryl group
- $X = O, C, N$

— Ethylene bridge

— Amine group

— Substitution on aryl group →

- Diaryl substitution is essential for activity.
- Both alkyl group can be phenyl or any other heterocyclic ring which also increase activity.
- Ar' may be methyl aryl group which show activity.



Diphenhydramine

Doxylamine

activity ↑

- Substitution of G, R, O-CH3 group on aryl ring increase the activity.

eg.



Bromodiphenhydramine

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— Nature of [X]

- It can be substituted by three major elements.
- [X] = Nitrogen (N), Oxygen (O), Carbon (C).

- If X, substituted with any other element than activity of drug is less/lost.
- Now, they form three types of drugs.

i) When X = N (Nitrogen)



ii) When X = O (Oxygen)



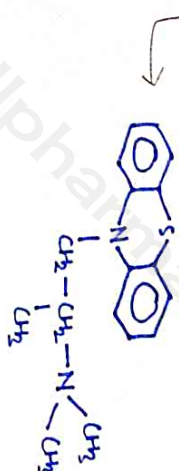
- Amino alcohol analogue

iii) When X = C (Carbon)



— Substitution on ethylene chain

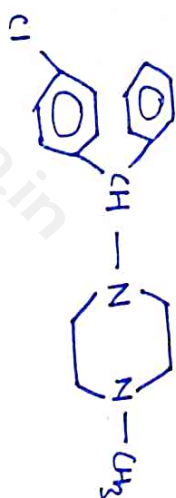
- Most of the antihistaminics have ethylene chain and which is essential for activity.
- Branching of this ethylene chain can decrease the activity.
- But promethazine is an exceptional case.



— Amino group

- 3^o amino group is essential for activity.
- The tertiary amino may be a part of heterocyclic ring which increase the activity multifold.

Eg. Chlorcyclizine

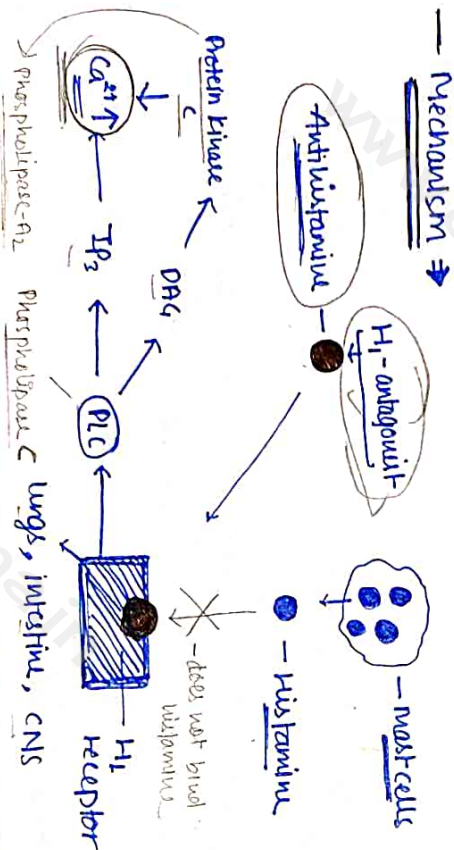


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H₁-Antagonists

- Those agents which blocks the H₁ receptor and inhibit the action of histamine.
- These are of two types:—
 - 1) first generation (sedative)
 - 11) Second generation
- 1) H₁ antagonists (first generation)
- These are competitive inhibitors of histamine at H₁-receptor.



- The histaminergic receptors are G-protein coupled type
- H₁ receptor are coupled to phospholipase-C and their activation leads to the formation of Inositol phospholipid (IP₃) and diacylglycerol (DAG)
- IP₃ → rapid release of Ca²⁺ from endoplasmic reticulum
- DAG → activates Protein kinase C.
- also activates phospholipase A₂.
- Now, H₁-antagonist bind to the H₁ receptor and block this receptor. So, histamine does not bind with receptor and not give any action.
- # H₁-antagonists (first generation) cause sedation due to more selectivity for CNS neurons. decrease the production of PLC
- Also due to their lipophilicity.
- functions →
 - Relaxation of smooth muscles - Bronchodilation.
 - Decrease vascular permeability
 - Pain, Pruritus, headache, hypotension → decrease

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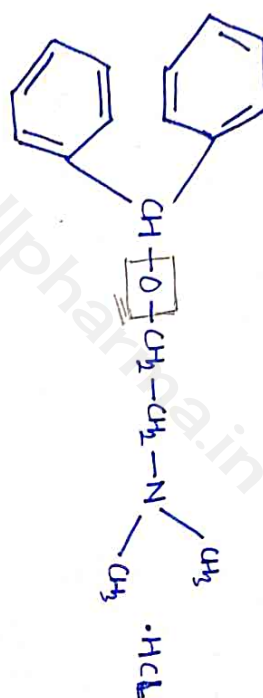
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• Various Drugs of first generation H_1 -Antagonists

• Synthesis →

1) Diphenhydramine Hydrochloride

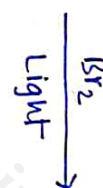
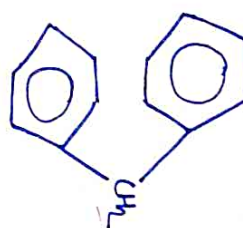


- It is a first generation antihistamines which is mainly used for treating allergies.

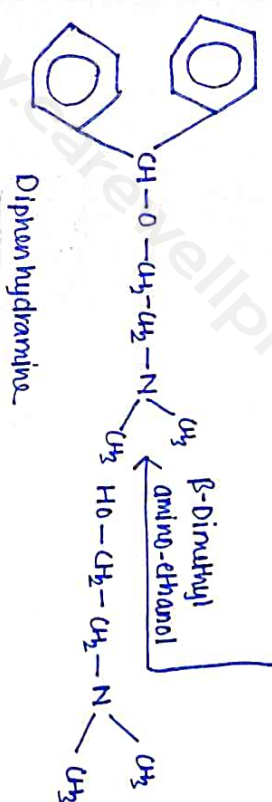
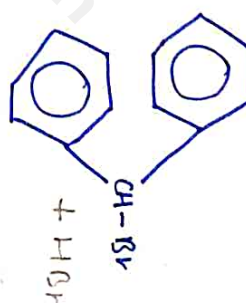
• M.O.A →

- It is a competitive antagonist of H_1 receptor.
- It also act on CNS (centrally) and cause Sedation, drowsiness.
- It also have some antimuscarinic activity.

Diphenyl methane



Diphenyl bromomethane



• Uses →

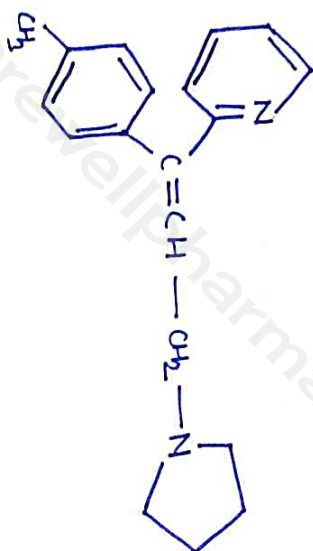
- Used to treat allergic condition like pain, pruritus, rhinitis, hay fever, common cold, nasus, watery eye, cough, runny nose & swelling etc.
- Easily cross BBB so cause sedation and used to relax and fall asleep (treat insomnia).

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- Used in prevention & treatment of motion sickness.
- also used in treatment of parkinson due to antimuscarinic activity.

ii) Triprolidine Hydrochloride



It is also a first generation H₁-antagonist.

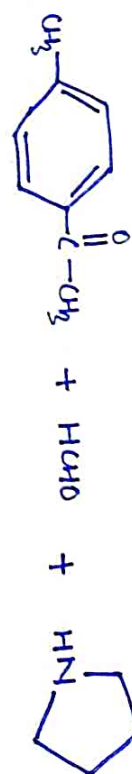
— MOA →

- Bind with H₁-receptor and block the receptor and also the action of histamine.

— Uses →

- Used to control the symptoms of histaminus.
- Combined with other cold drugs to provide relief in common cold.

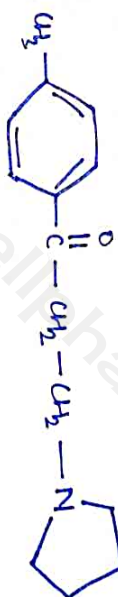
— Synthesis →



4-methylacetophenone

pyrrolidine

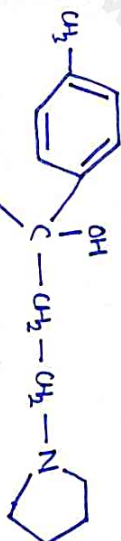
Manich reaction



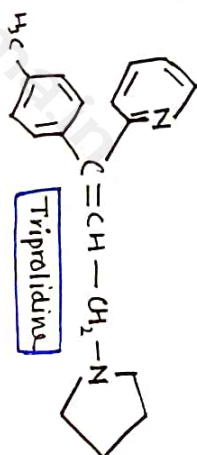
4'-methyl-3-pyrrolidino propiophenone



2-bromopyridine



— H₂O → Dehydration



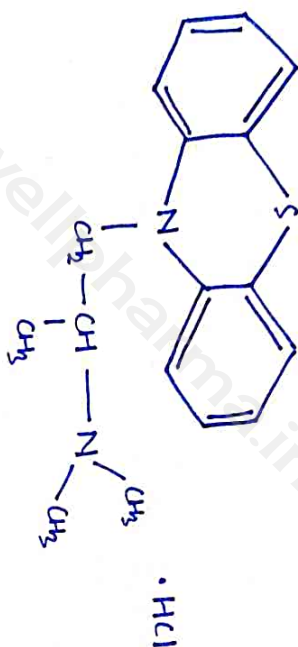
Triprolidine

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iii) Promethazine Hydrochloride

It is the hydrochloride salt form of promethazine. It having antihistaminic, sedative and antiemetic properties.



— Mechanism of action →

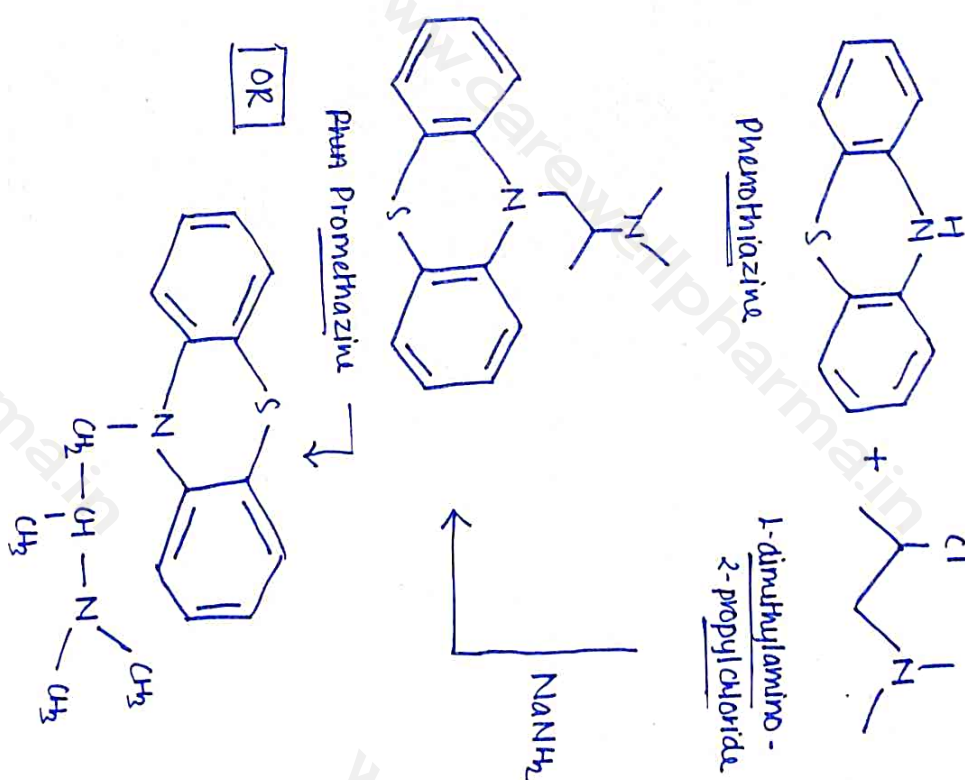
- It act as a H_1 receptor antagonists.
- It also inhibit the central histaminergic receptors (dopamine, serotonin).
- It also act as anticholinergic (moderate affinity)

— Uses →

- Used as sedative for treatment of insomnia.
- Used for medication of allergy like rhinitis and other allergic reactions.

Synthesis of promethazine

Alkylation of promethazine with 1-dimethylamino-2-propylchloride give promethazine.



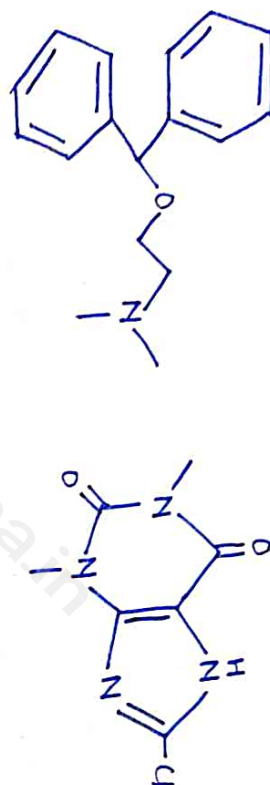
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4) Dimenhydrinate $\Rightarrow$



$[53-55.5\%]$ — chemical composition — $[44-47\%]$

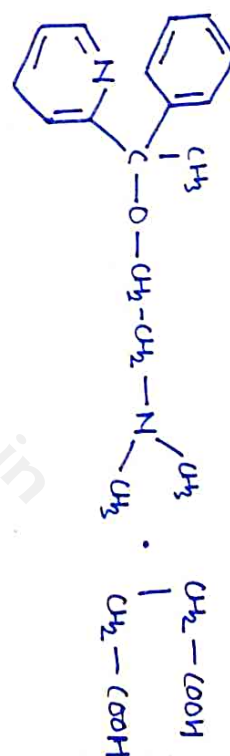
• MOA $\rightarrow$

- The antiemetic property is due to H_1 antagonism in the vestibular system in the brain.

• Uses $\rightarrow$

- Used to treat nausea, vomiting & prevent motion sickness.
- helps in treatment of ear congestion
- relieve vestibular disorder.

5) Doxylamine succinate $\Rightarrow$



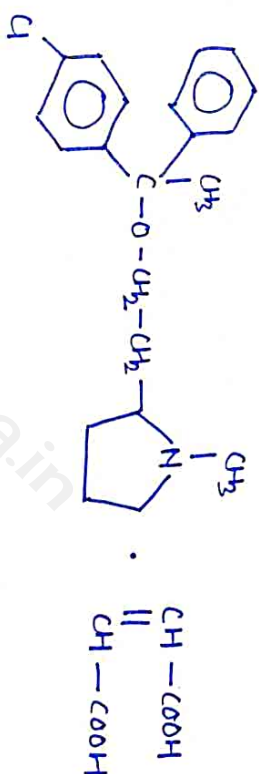
• MOA $\rightarrow$

- H_1 antagonists (same).

• Uses $\rightarrow$

- treat allergic symptoms allergy, hay fever & common cold
- used to treat insomnia.
- prevent morning sickness in pregnant woman in combination with Vitamin B_6 (pyridoxine).

6) Clemastine fumarate $\rightarrow$



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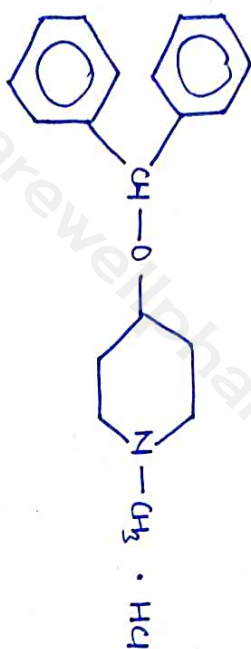
• MOA →

- H_1 Antagonists (same as others).

• Uses →

- treat allergic symptoms (same as others).
- it also have some anticholinergic activity.

7) Diphenylpyraline hydrochloride ⇒



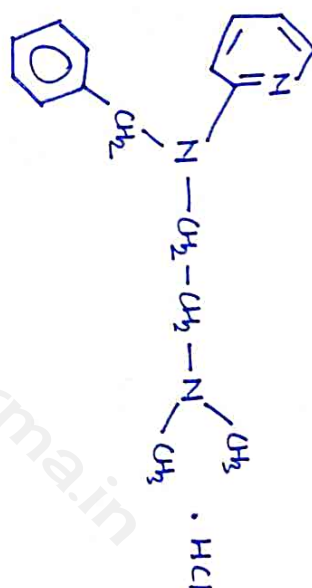
• MOA →

- competitive antagonists for H_1 receptor, after binding on H_1 receptor it suppress the activity of histamine (cause temporary relief).

• Uses →

- Same as others.

8) Tripehenamine hydrochloride →



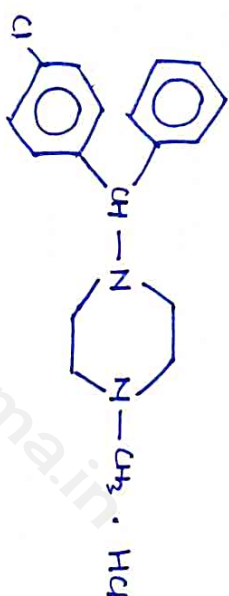
• MOA →

- H_1 Antagonists (same as others)
- also have little anticholinergic activity.

• Uses →

- used in treatment of upper respiratory tract allergic condition like asthma, hay fever, rhinitis.
- other allergic symptoms treatment.

9) chlorcyclizine hydrochloride ⇒



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- H_1 Antagonists (same as other).

- treat allergic symptoms like rhinitis, urticaria and pruritus.
- used for treating hepatitis C

Clc1ccc(cc1)C2=CC=CC=C2CN3CCN(CC4=CC=C(C)C=C4)CC3

- H_1 antagonists (same as other)
- also act as dopamine antagonists & anticholinergic.

- used for treating of nausea, motion sickness
- or vertigo.
- safely used in the treatment of nausea in

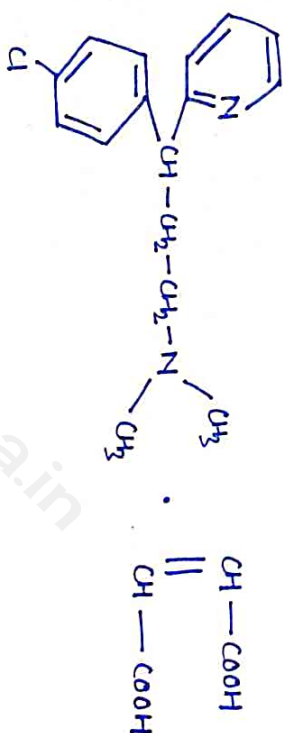
ii) Bucilzina hydrochloride →

CC(C)(C)c1ccc(cc1)CN2CCN(CC2)c3ccccc3 + Clc1ccccc1 $\xrightarrow{\text{NaOH}}$ CC(C)(C)c1ccc(cc1)CN2CCN(CC2)c3ccccc3c4ccccc4Cl

- Blocks H_1 receptor in vomiting centre.
- also block muscarinic receptors.

- used as an anti-vertigo or antiemetic agent
- also act as CNS depressant.

12) Chlorpheniramine maleate $\Rightarrow$



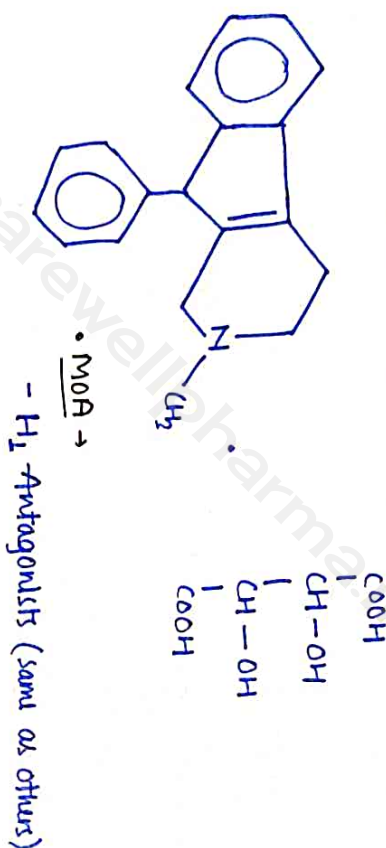
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- MOA →
- H_1 antagonists (same as others)

- Uses →
- relieving allergic symptoms (same as others).

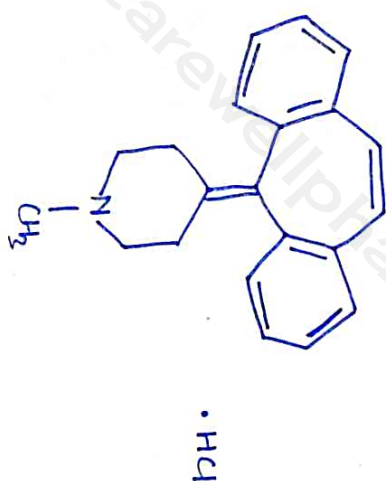
13) Phenidamine tartrate →



- MOA →
- same (H_1 antagonists)

- Uses →
- treat allergic symptoms (same as others)
- act as sedative, hypnotics & antiemetic
- for motion sickness.

15) Cyproheptadine hydrochloride →

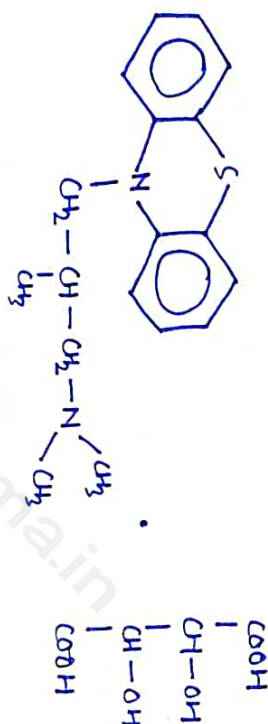


- MOA →

- potent H_1 antagonists
- also have anticholinergic, antidiarrhoeal and anti serotonergic activity.

- Uses →

14) Trimiprazine tartrate → Alimemazine

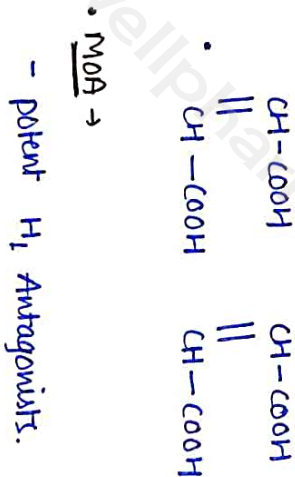
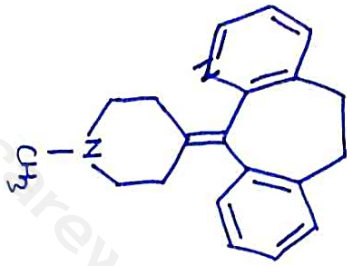


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- treat allergic reactions
- treatment of akathisia (movement disorder).
- used as an antiemetic in cyclic vomiting syndrome.

16) Thizatidine Maleate $\Rightarrow$



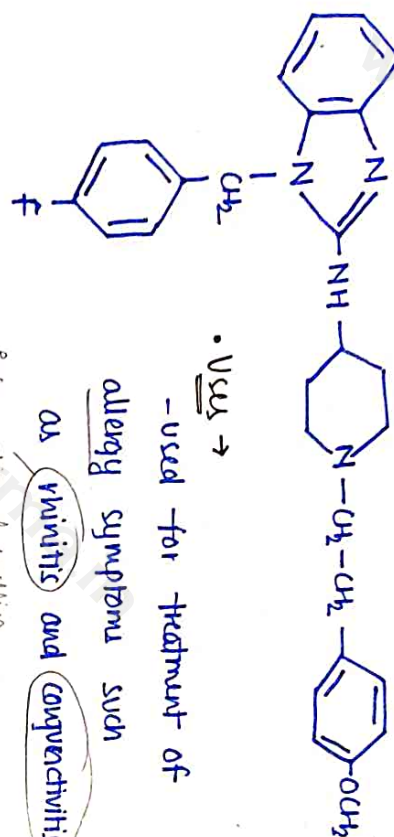
- Uses $\rightarrow$
- treating the symptoms of upper respiratory mucosal congestion in perennial and allergic rhinitis.
- relief of nasal congestion and eustachian tube congestion.

11) H_1 Antagonists [second Generation]

- Major side effect of first generation antagonist is sedation, drowsiness because they easily cross the BBB.
- But these second generation H_1 -antagonists does not cross BBB, so it does not cause sedation.
- They are more selective for peripheral H_1 receptors.
- They are hydrophilic in nature.
- Mechanism of use are same as first generation but does not give their action on CNS.
- Various drugs $\rightarrow$
- 1) Festemizole
- MOA $\rightarrow$
- Reversible blockage of H_1 receptor in GIT, Uterus, large blood vessels & bronchial muscles.

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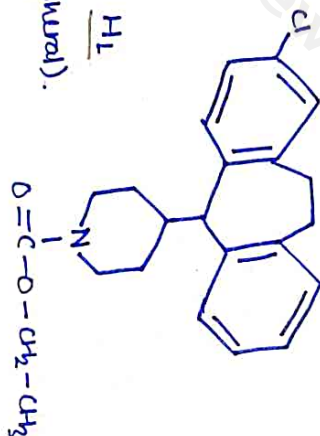
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- Uses →
- used for treatment of allergy symptoms such as vinitis and conjunctivitis.
inflammation & swelling of mucous membrane of nose

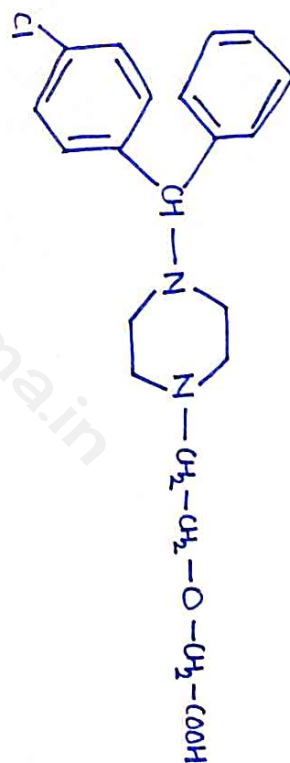
2) Loratadine →

- MOA →
- competitively antagonist of H₁ receptor (peripheral).



- Uses →
- used alone or in combination with pseudoephedrine sulphate for relief of seasonal allergic such as rhinitis, pruritus and erythema.

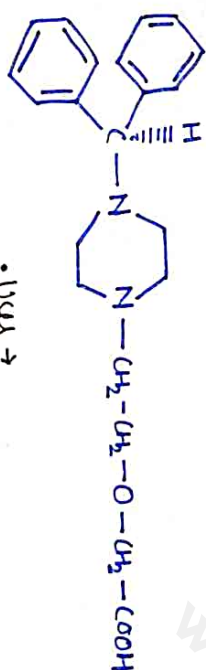
3) Cetirizine →



- MOA →
- acts as H₁ receptor antagonist. (peripheral).

- Uses →
- used to relieve allergy symptoms like runny nose, itching eye, sneezing, itching etc..

4) Levocetirizine → levorotatory enantiomers of cetirizine.



- MOA →
- active enantiomers of cetirizine which blocks H₁ receptor.
- Uses →
- similar as cetirizine but more potent.

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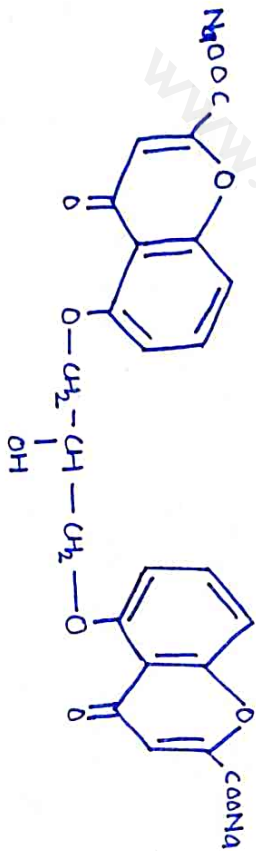
iii) Mast cell stabilizers

It inhibits the release of histamine, leukotrienes and other substances that cause hypersensitivity reactions from the mast cells.

- It is a mast cell stabiliser and it has anti-inflammatory activity.

↳ inhibit the release of prostaglandins and leukotrienes and inhibit activation of eosinophils, neutrophils, monocytes & platelets.

1) Cromolyn Sodium →



- MOA →
- It inhibits degranulation of mast cells thus preventing the release of histamine and leukotrienes.

- It acts by inhibiting calcium influx.

• Uses →

- used for the management of bronchial asthma
- It's nasal solution is used for allergic rhinitis.
- It's eye drop are used to treat allergic conjunctivitis - inflammation of outer membrane of eyeball and inner eyelid.

H₂ - Antagonists

- These are those agents or drugs which block the H₂-receptor which present in the stomach.
- These drugs mostly decrease the production of acid in stomach
- these drugs are used in the treatment of gastric and duodenal ulcer.

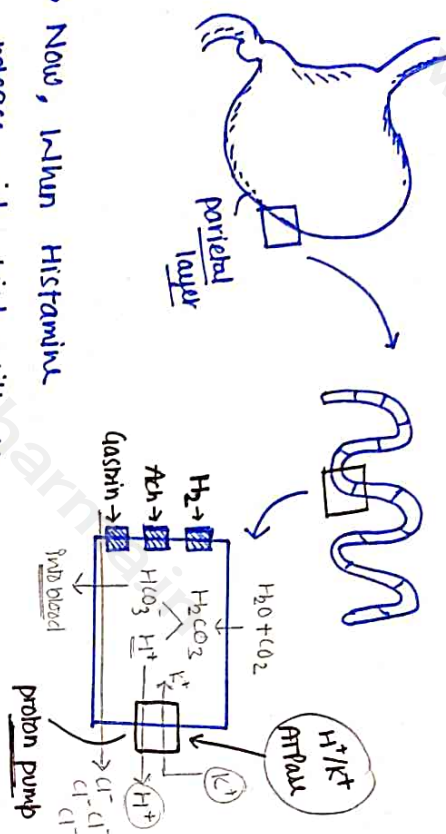
— Mechanism →

- H₂ receptor are present in the parietal cells in the parietal layer of stomach.

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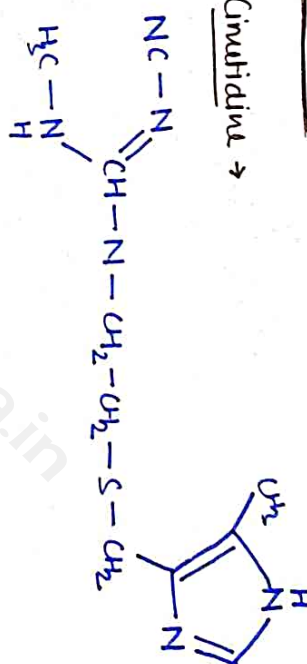
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- Now, when Histamine release, it bind with H_2 receptor
- Now, H_2 receptor are also GPCR, so it activates the **cAMP**
- Now this cAMP increase **$[Ca^{2+}]$** which further increase or activate proton pump and increase the secretion of H^+ — exocytosis
- Which further increase the production of HCl.
- * Now, H_2 Antihistamine block the H_2 receptor so all the mechanism inhibit. and all the action of Histamine reduced.
- HCl decrease ↓.

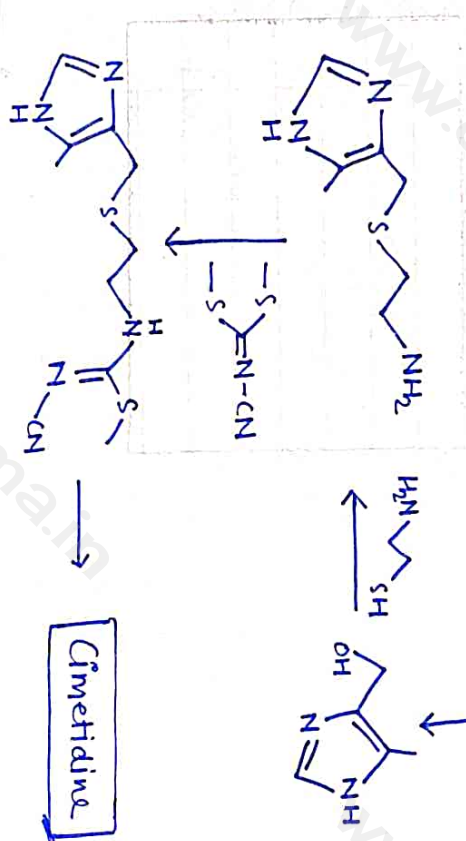
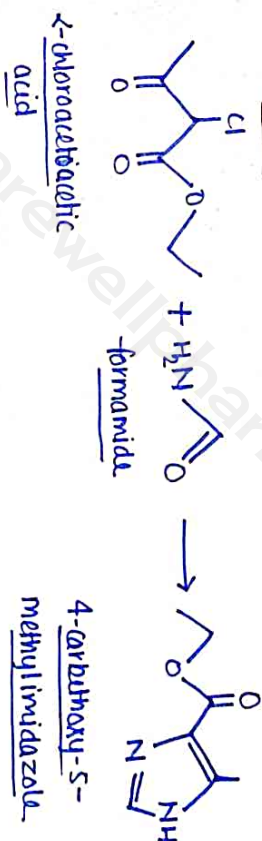


• Various Drugs →

1) Cimetidine →



— Synthesis →



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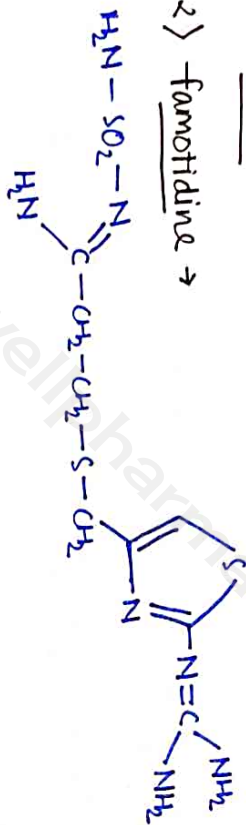
— MOA →

- Antagonist of H_2 -receptor and decrease the activity of histamine.

— Uses →

- Used in treatment of gastric and duodenal ulcers.

2) famotidine →



• MOA →

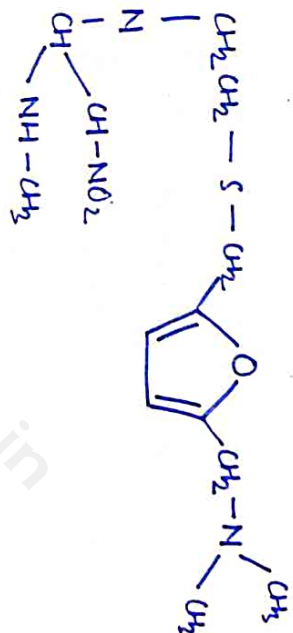
- more potent competitive inhibitors of histamine H_2 receptor than cimetidine.

• Uses →

- used in treatment of gastric and duodenal ulcers.

3) Ranitidine →

- MOA →
- Same as others.



• Uses →

- same as others.

GASTRIN PROTON PUMP INHIBITORS (PPI)

- These are the irreversible inhibitors of gastric parietal cells proton pump.

- used to control the gastric acidity and duodenal ulcer.

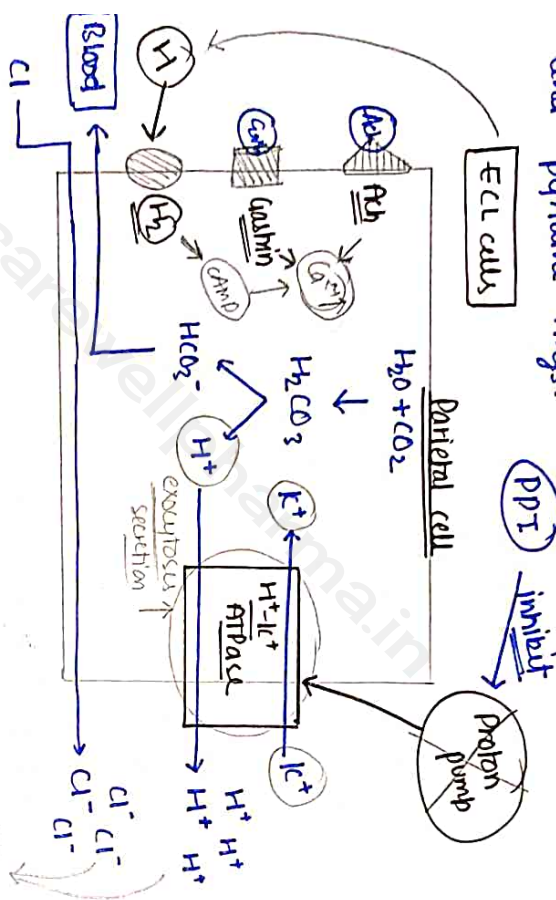
— mechanism →

- There are one enzyme H^+/K^+ ATPase which present in parietal cell of stomach and it is responsible for HCl secretion.

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- Now these drugs contain a sulphonyl group in a bridge between substituted benzimidazole and pyridine rings.

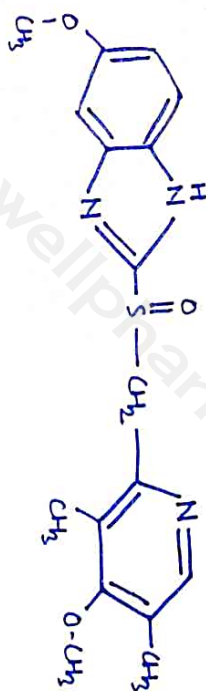


Gastrin hormone produced by G-cell located in pyloric gland which act on gastrin receptor.

- Now, when PPI introduce ^{from blood} into parietal cells its sulphonyl group bind with H^+-K^+ ATPase-SH enzyme then they form Drug enzyme complex, which further metabolize easily & irreversibly inactivate the proton pump.

- Now, H^+ ion does not secrete out so production of HCl decreases.
- It blocks gastric acid secretion even if Histamine bind with their receptor.

1) Omeprazole →



• MOA →

- Selective and irreversible proton pump inhibitors.
- It suppresses gastric acid secretion by inhibiting the H^+-K^+ ATPase in gastric parietal cells.

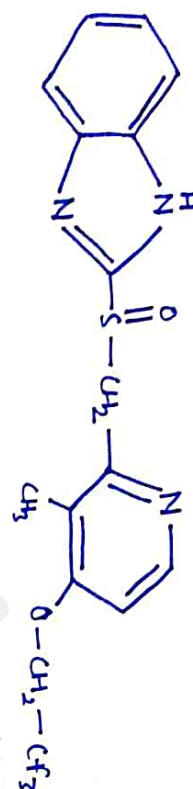
• Uses →

- It blocks final step of acid production
- used in the treatment of peptic ulcer, erosive esophagitis and Elison Syndrome, drug-induced to esophagitis by the stomach acid.

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2) Lansoprazole $\Rightarrow$



• MOA $\rightarrow$

- Same as other.

• Uses $\rightarrow$

- Used in treatment of duodenal and gastric ulcer.

- Zollinger-Ellison syndrome (ZE syndrome).

(in which tumors cause the stomach to produce too much acid, resulting in peptic ulcers).

- treatment of

gastroesophageal reflux

Stomach content rise up into esophagus.

which cause heart burn, chest pain,

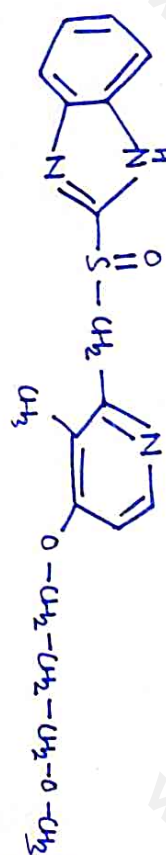
vomiting etc.

3) Rabeprazole $\Rightarrow$

• MOA $\rightarrow$

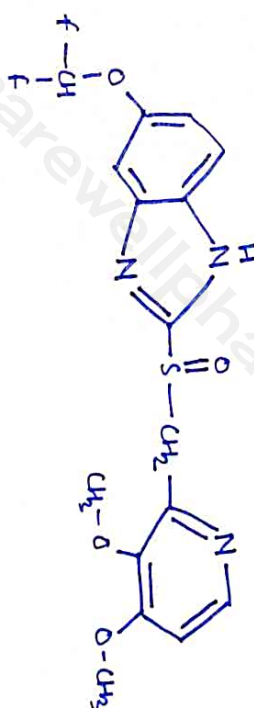
- Same as other.

4) Pantoprazole $\Rightarrow$



• Uses $\rightarrow$

- Same as omeprazole & lansoprazole.



• MOA & Uses $\rightarrow$

- Same as other.

COMPLETE

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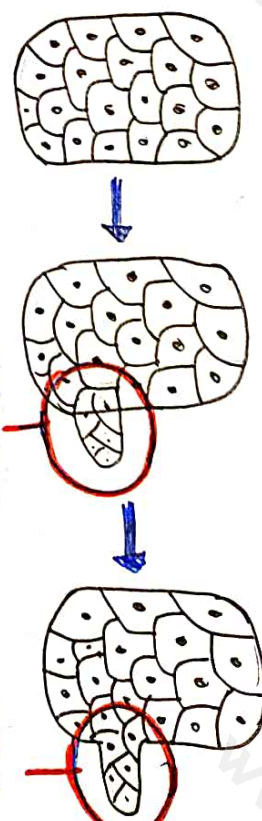
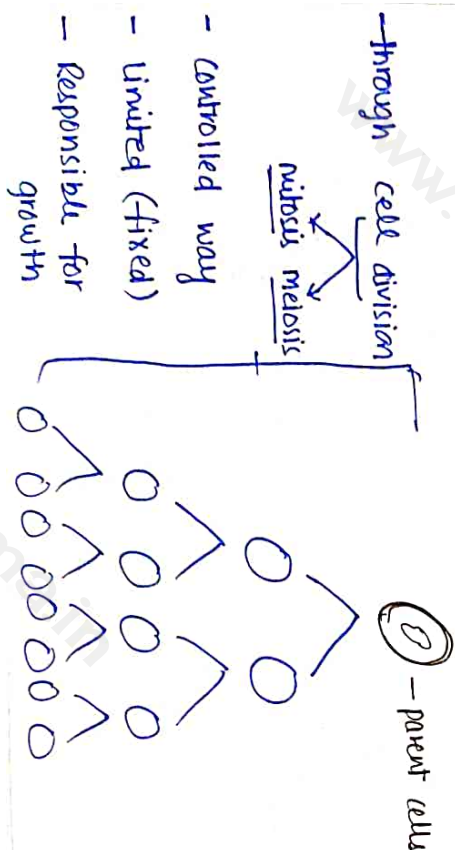
ANTI-NEOPLASTIC AGENTS

These are those agents or drugs which are used in the treatment of cancer.

Anti-neoplastic agents
oppose neoplasm drugs

CANCER → tumor → cancer

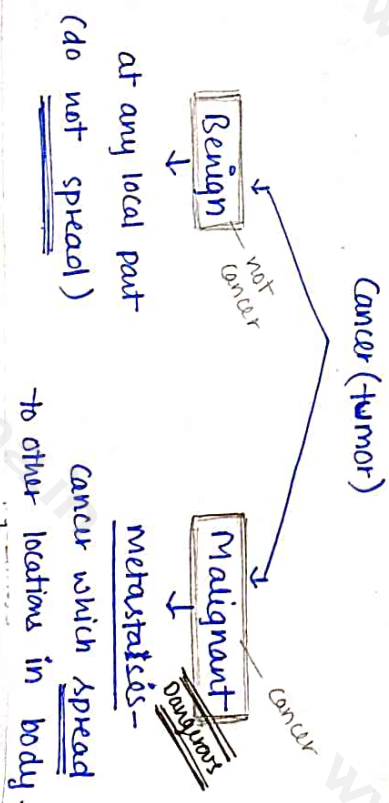
It is a group of disease which involve an abnormal and uncontrolled cell division in body cells.



• Neoplasm — An abnormal mass of tissue that forms when cells grow and divide more than they should or do not die when they should.

— This leads to the formation of tumor.

• On the basis of Nature, It can be —



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In condition of cancer

The diagram illustrates the progression of cancer. It begins with a single cell on the right, which divides into two cells. These two cells then divide into four cells, and so on, forming a branching tree structure. An arrow points from this branching structure to a large, dense, irregular mass of cells on the left, labeled "cells". Another arrow points from this mass to a box labeled "Uncontrolled cell division", which then points to a box labeled "Cancer".

- Reason →

- UV rays

- Radiation -
- Heredity -

proto-oncogenes

Oncosuppressor genes
 Onco stimulated genes

- Diet (tobacco, etc...)

er-

धातु, गुण, पात्र, इत्यादि, कर्तृक इत्यादि

It can be classified as many ways, it is mainly based on where it begins —

- 1) Carcinomas - If begins in the skin or the tissue that covers the surface of internal organs and glands.

eg. Breast cancer, prostate cancer, lung cancer
and colorectal cancer etc--

ii) Sarcomas - a sarcoma begins in the tissue that supports and connects the body.

A sarcoma can develop in fat, muscles, nerves, tendons, joints, blood vessels, lymph vessels, cartilage or bone.

iii) Leukemias - Blood cancer.

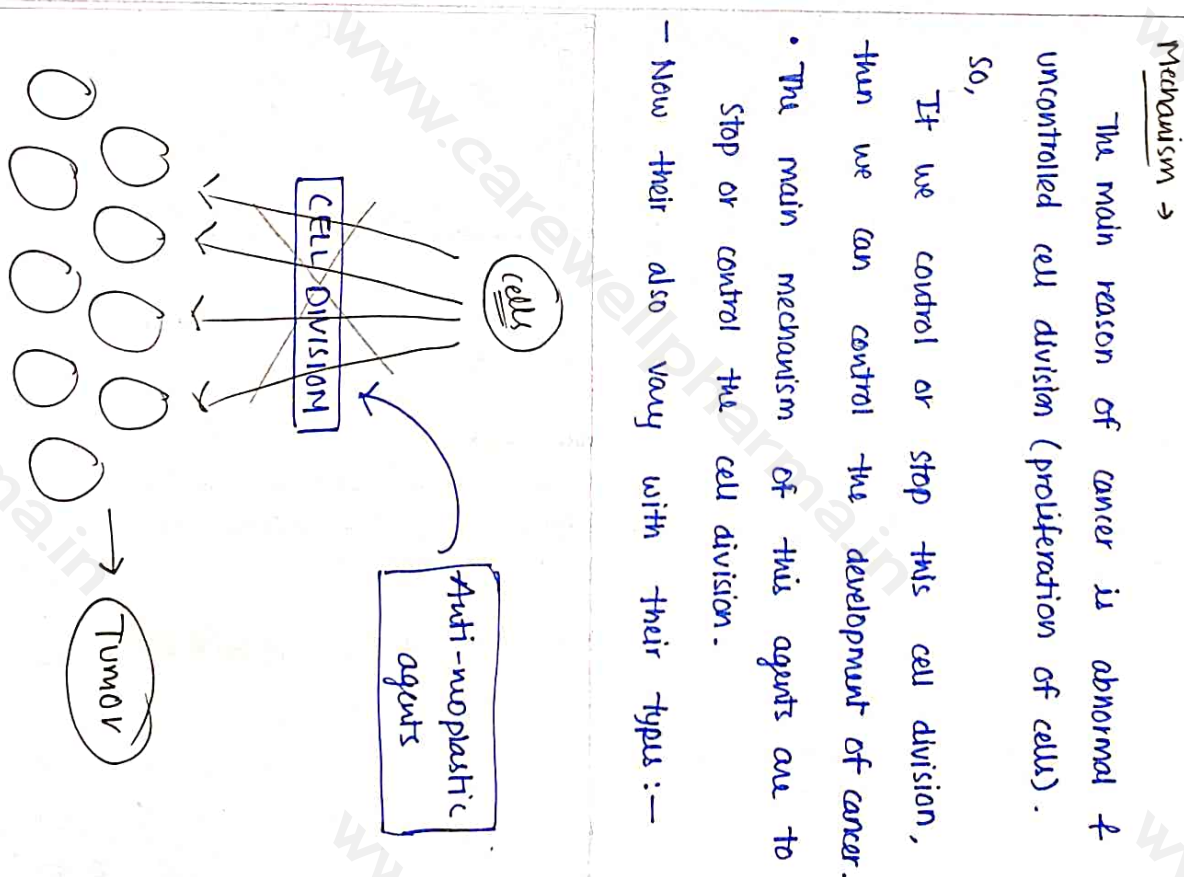
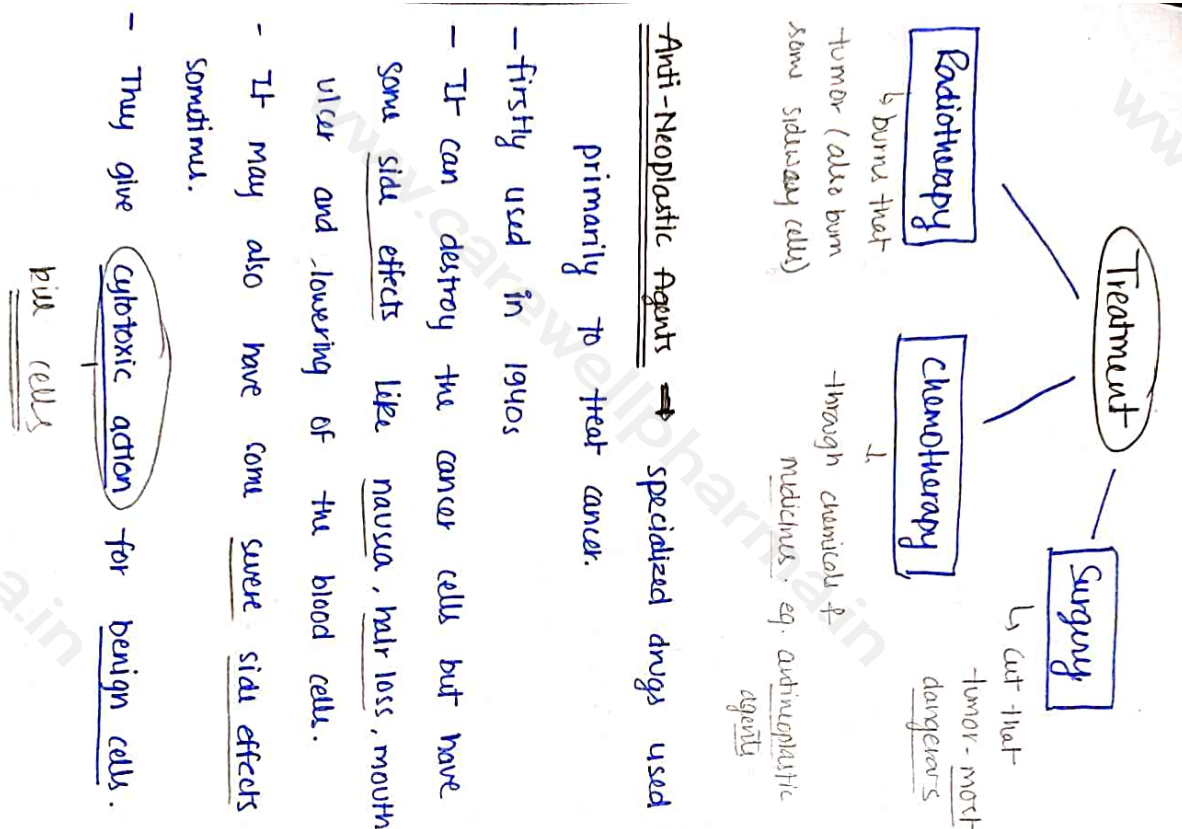
It begins when healthy blood cells change and grow uncontrollably.

iv) Lymphomas - It begins in the lymphatic systems.

helps to fight against infectious

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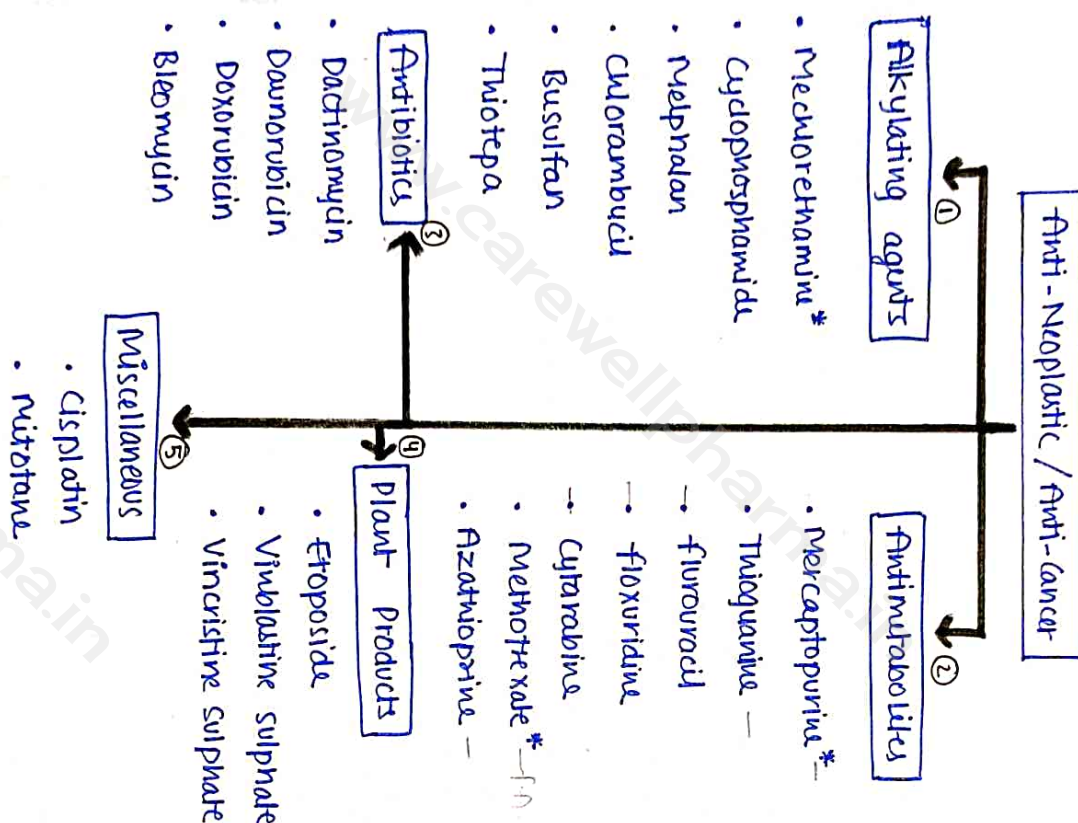
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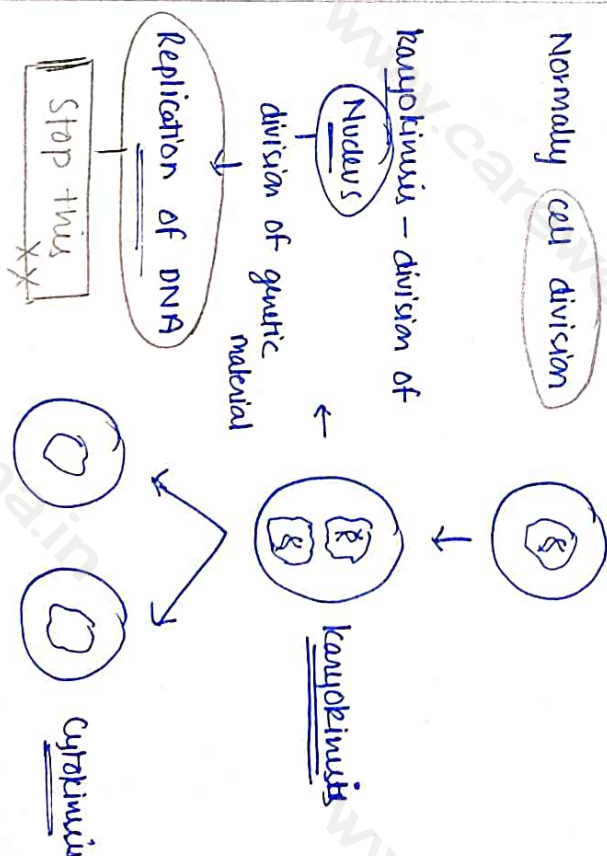
Classification →



① Alkylating Agents

These agents which comes under the class of antineoplastic or anticancer drugs. These drugs binds covalently with the structure of DNA which inhibit the replication of DNA resulting in the decrease of cell division.

— Mechanism ⇒



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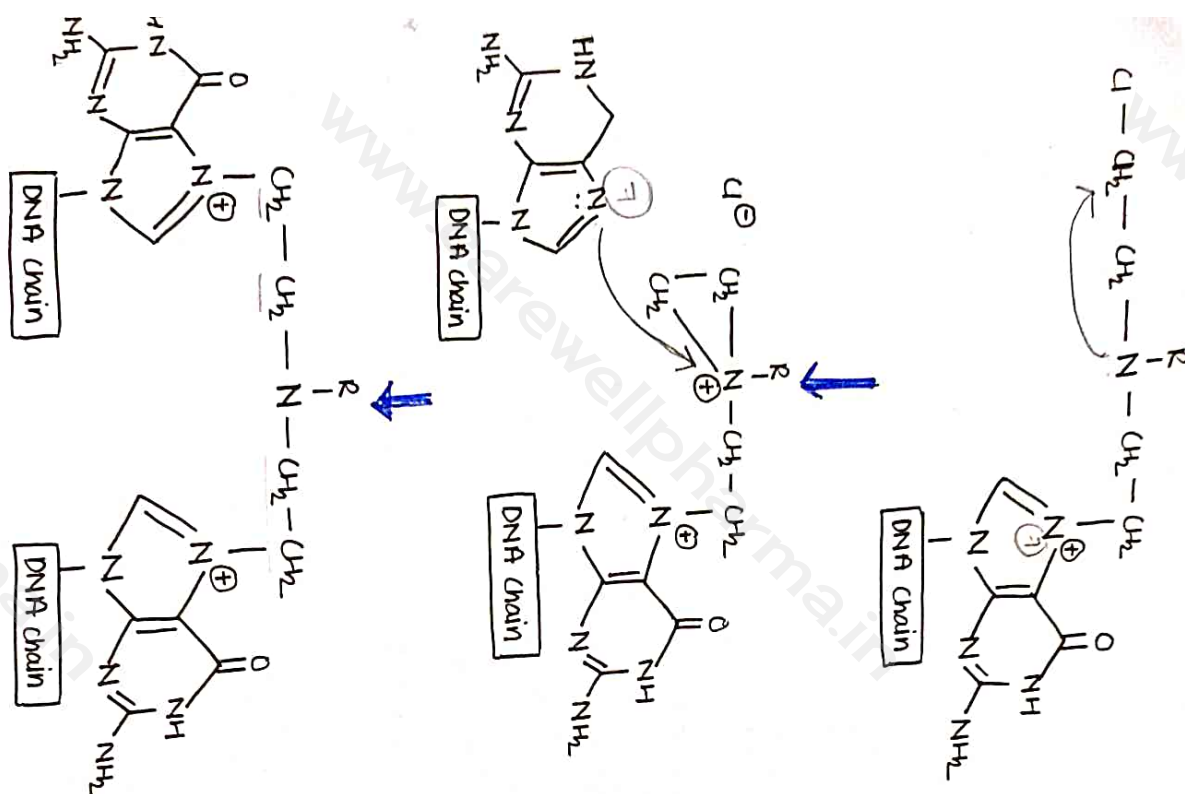
- Carbonium ion

- DNA chain

- Now this alkylating agent again bind with 7th position of guanine (DNA), on their second side.

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- This results in the cross-linking of two nucleic acid chains, abnormal base pairing & scission of DNA strand. Nitrogenous base
 - Mismatch the pair of Nitrogenous base pair in DNA. → Inhibition of DNA synthesis.
- ↓
- Decrease cell division / cell death
-
- Various Drugs ⇒
 - ① Medroretamine (mustin) →
- $$\text{CH}_3-\text{N} \begin{array}{l} \text{CH}_2-\text{CH}_2-\text{Cl} \\ \text{CH}_2-\text{CH}_2-\text{Cl} \end{array}$$
-
- Mechanism →
- It comes under the class of Nitrogen mustard.
 - It damages DNA via the formation of cross link (prevent DNA synthesis).
 - Induction of mispairing of the nucleotides leading to mutations.

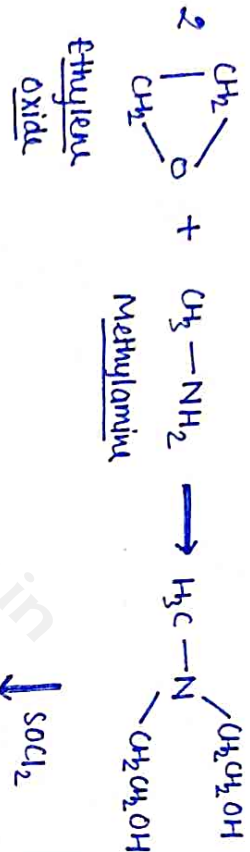
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Synthesis →

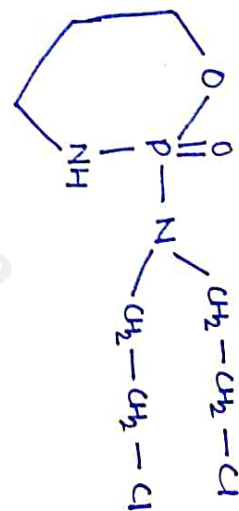


In this synthesis, Ethylene oxide react with methylamine to give 2,2-(methylimino) diethanol which further on chlorinated with SOCl_2 gives mustin [OR] methlorethamine.

Uses →

- Used in the treatment of Hodgkin's disease.
 - also used for the treatment of metastatic carcinoma.
 - used topically in the treatment of mycosis fungoides (T-cell lymphoma).
- Cancer of part of immune system.
Lymphatic system.
Most common type of blood cancer.

② Cyclophosphamide →



• MOA →

- The active metabolite of cyclophosphamide i.e. phosphoramide mustard forms DNA cross links between and within DNA strands at guanine N-7 position.

- This is irreversible and leads to cell death.

• Uses →

- used in the treatment of lymphomas, some solid tumors and some forms of leukaemia.
 - It is also active against Hodgkin's disease, breast, ovarian cancer and lung cancer.
 - also used in some various non-neoplastic disease etc...
- Blood cancer.

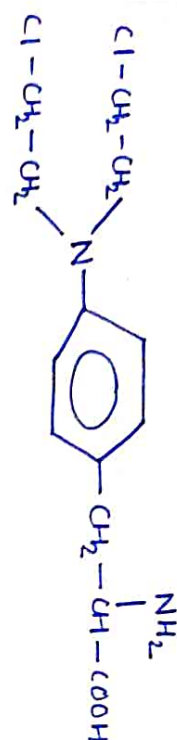
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③ Melphalan →



• MOA →

– Same as other

• Uses →

– used to treat retinoblastoma (retina cancer).

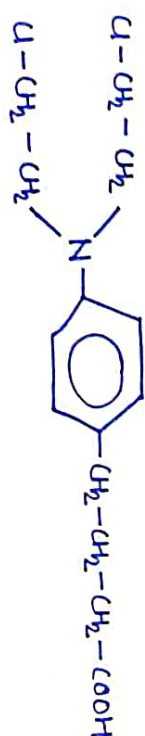
– used with prednisone in the treatment of amyloidosis.

related to a type of bone marrow cancer called myeloma/multiple myeloma.

– used mainly in the treatment of multiple myeloma and carcinoma of breast and ovary.

A cancer of plasma cells overproduction

④ Chlorambucil →



• MOA →

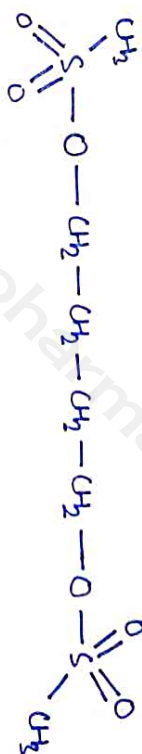
– Same as other

• Uses →

– used in the treatment of chronic lymphatic leukemia, malignant lymphomas.

– used as an immunosuppressive agent for various autoimmune disorders.

⑤ Busulphan →



• MOA →

– It is a bifunctional alkylating agent.

– It interacts with the thiol groups of proteins and nucleic acids and forms

DNA-protein and DNA-DNA cross links.

– these cross-linkages prevents the synthesis and function of DNA.

• Uses →

– Used to treat chronic granulocytic leukaemia.

– also used as immunosuppressive effect on bone marrow.

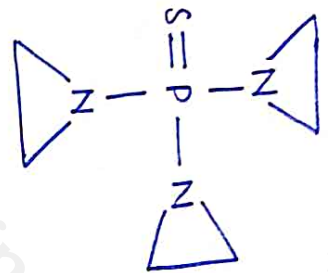
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⑥ Thiopeta →

• MOA →

It acts on 7th position of



guanine base of DNA which cause cross-linking with DNA double helix strands which further decrease/stop cell division.

• Uses →

- used for treating breast, ovarian and bladder cancer.
- also used in the treatment of various lymphomas.

⑦

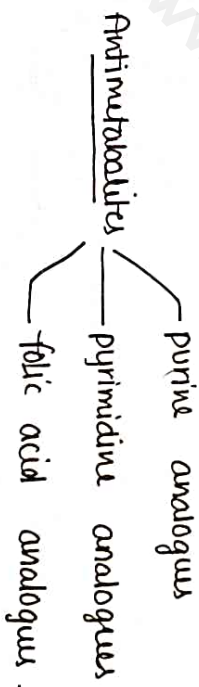
Antimetabolites →

Metabolites are those important constituents which are responsible for the synthesis of DNA.

[eg]. Purine, pyrimidine, folic acid which are used in the synthesis of DNA.

• Antimetabolite is a substance that replaces or inhibit a specific metabolite of a cell which interfere with normal cellular metabolic functions.

• Antimetabolite has similar structure as metabolite which prevent the biosynthesis of DNA.

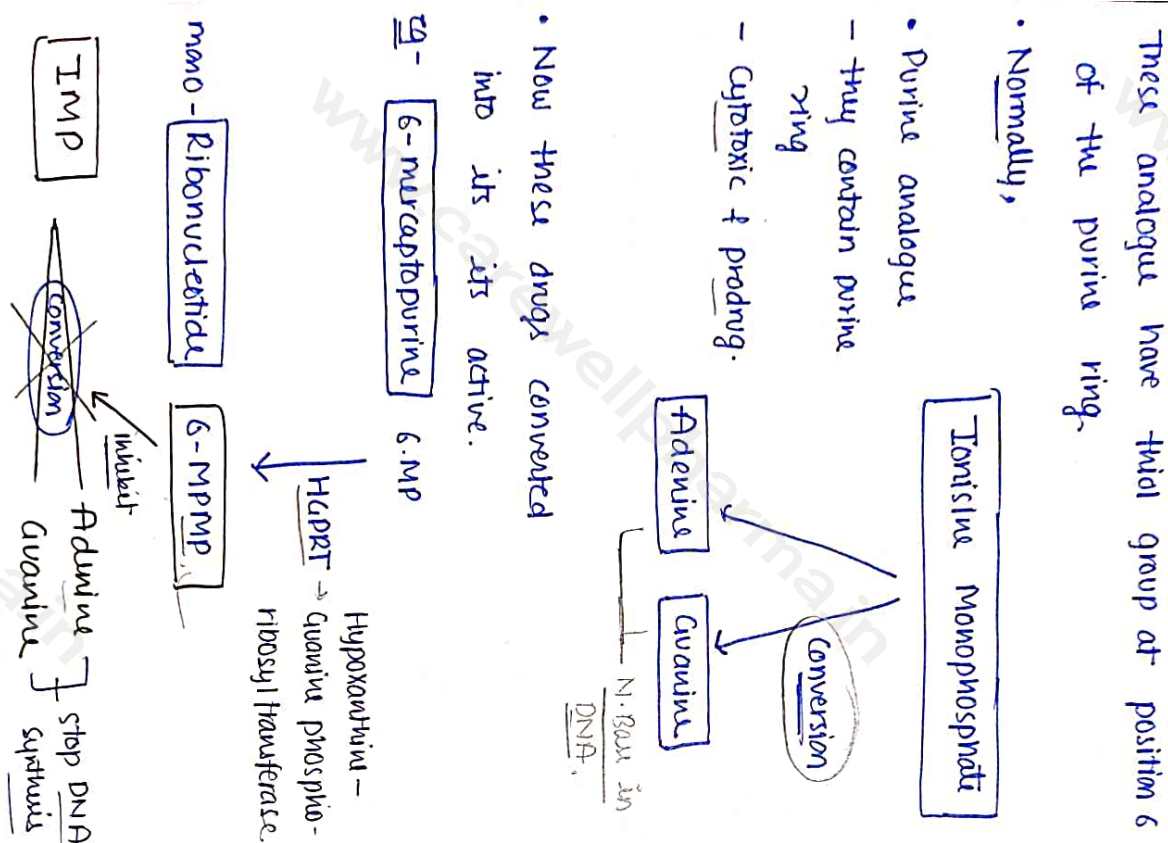


1) purine analogues →

Mercaptopurine, Thioguanine, Azathioprine

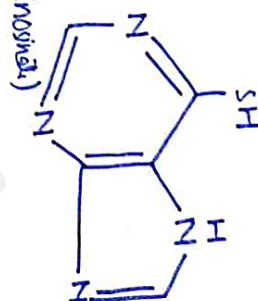
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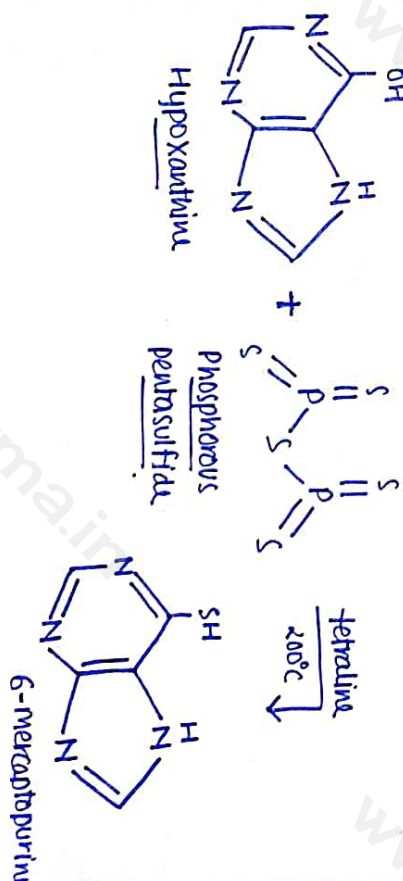
① Mercaptopurine →

- MOA →
- It metabolised into 6-MPMP (6-thioinosine) by HGPRT
- Inhibits conversion of inosine acid to adenylic acid and xanthylic acid
- prevent the purine biosynthesis.



- Uses →
- used in the treatment of acute monocytic leukaemia.

• Synthesis →



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② Thioguanine →

• MOA →

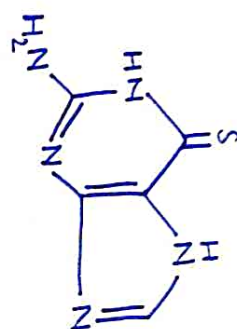
- same as 6MP

• Uses →

- used in the treatment of acute

non-lymphocytic leukaemia

- type of cancer of blood and bone marrow with excessive immature white blood cells.



③ Flazathioprine →

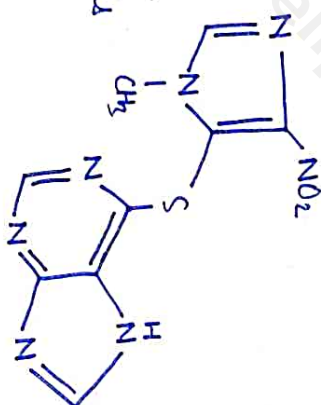
• MOA →

- It antagonizes

purine metabolism.

- inhibit synthesis

of DNA, RNA + proteins.



• Uses →

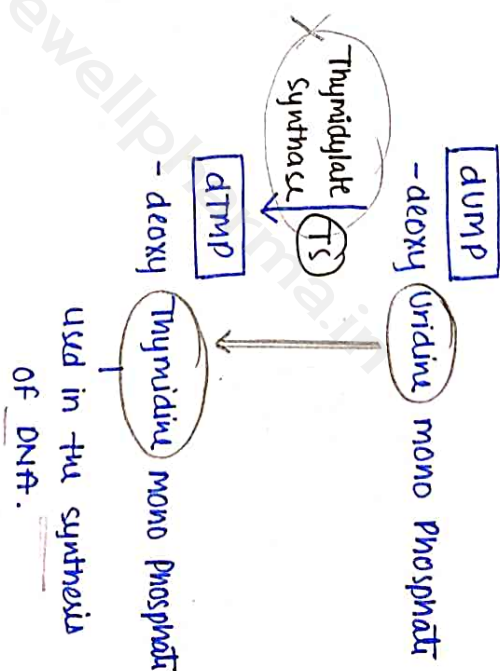
- used as an immuno-suppressant.

- used for preventing renal transplant rejection.

① Pyrimidine analogues →

- fluorouracil, flouxuridine and cytarabine.

Normally,



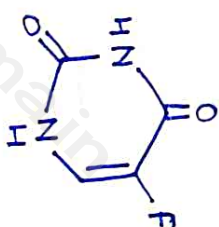
• Now, these pyrimidine analogues inhibit

this enzyme (TS) Thymidylate synthase which

further inhibit the synthesis of thymidine.

- which further inhibit DNA & cell division.

① fluorouracil →



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• MoA →
- inhibit enzyme Thymidylate synthase.
(same as above).

• Uses →
- Used for treatment of oesophagus, and, stomach, pancreatic, breast and skin cancers.

② flouxuridine →

• MoA →
- similar as fluoro uracil

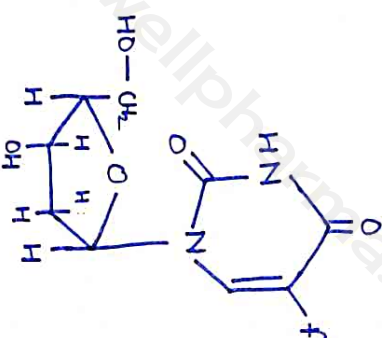
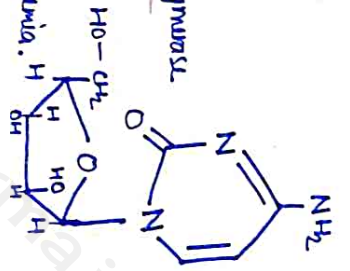
• Uses →
- used for treatment of kidney and stomach cancer.

- Used in treatment of colorectal cancer.

③ Cytarabine →

• MoA →
- inhibit DNA polymerase.

• Uses →
- used in acute non-lymphocytic leukaemia.

iii) folic acid analogues →
- Methotrexate

Normally,

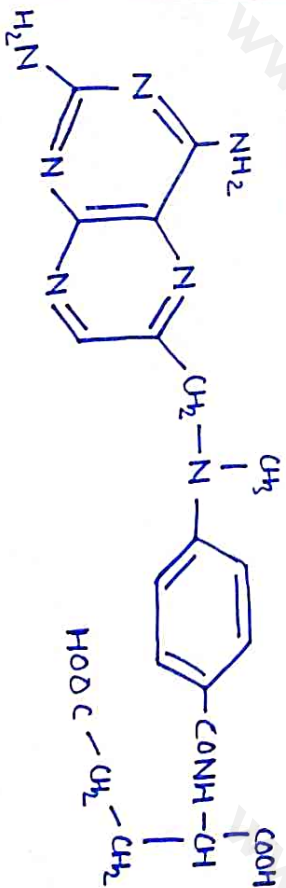
• folic acid analogues }
(methotrexate)

Inhibit the enzyme

folic acid → Dihydrofolate Reductase → Tetrahydro folic acid (THF)

necessary for the synthesis of purines, thymidine & some amino acids.

① Methotrexate →

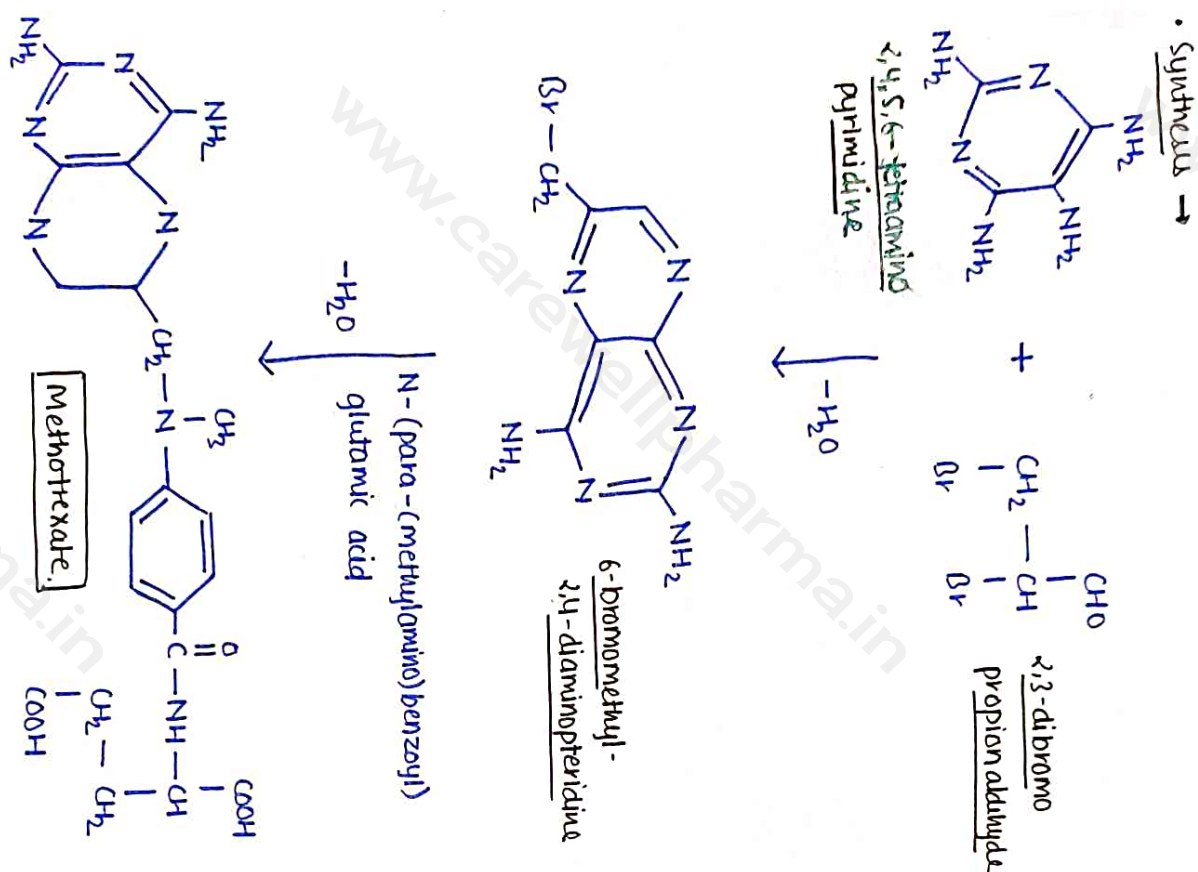


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- MOA →
- inhibits the enzyme dihydrofolate reductase thus prevent the formation of THF.
- Uses →
- used to treat psoriasis, acute lymphoblastic leukaemia
 - also used as an immunosuppressant.
 - also used to treat breast, neck and lung cancer.
- ③ Antibiotics →
- Antibiotics recently recognised as an important class of antineoplastic agents.
- mechanism →
- Antibiotics act by binding to DNA and intercalation & strand breakage
- fitting into the helical lattice b/w specific bases
- ↓
- inhibit cell replication
- ↓
- Blocking the transcription of RNA & DNA

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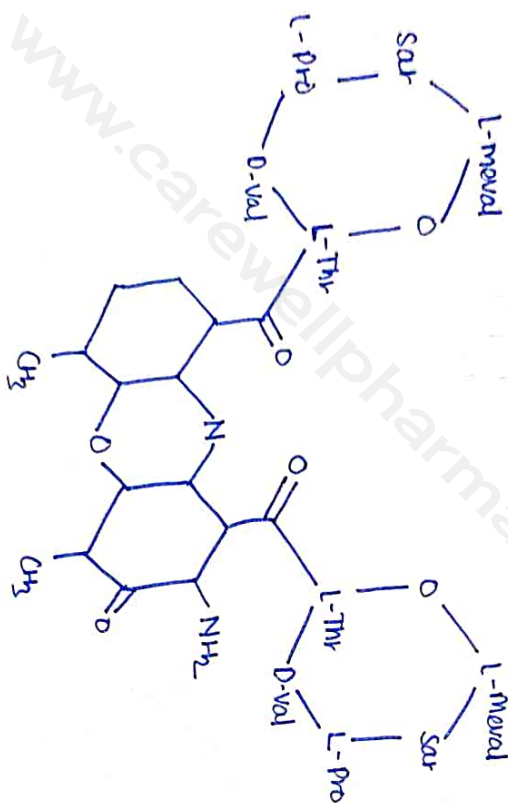
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- Many antineoplastic antibiotics are obtained from fungus streptomycetes include bleomycin etc..

— Various drugs →

① Dactinomycin →



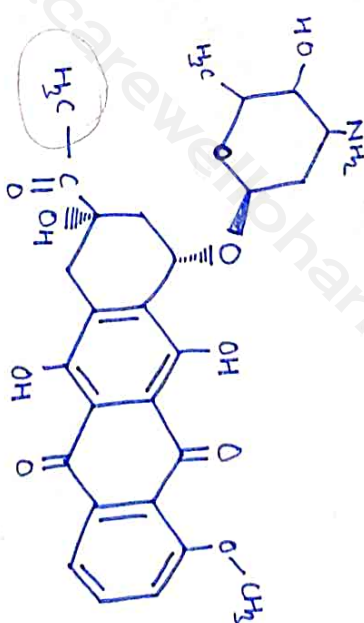
- MOA →
- Dactinomycin binds to DNA and blocks the RNA transcription.
- Decline protein synthesis.

• Uses →

- Used in treatment of tumours of testis and uterus, Kaposi's sarcoma, osteogenic sarcoma and other solid tumors.

② Daunorubicin → (daunomycin)

It comes under the class of anthracycline
- obtained from streptomycetes penicillatus



- MOA →
- It interacts with DNA by intercalation and inhibit the macromolecular biosynthesis and inhibit the topoisomerase II which relax super coils in DNA transcription.
- Uses →
- Used in treatment of leukaemia, Kaposi's sarcoma etc.

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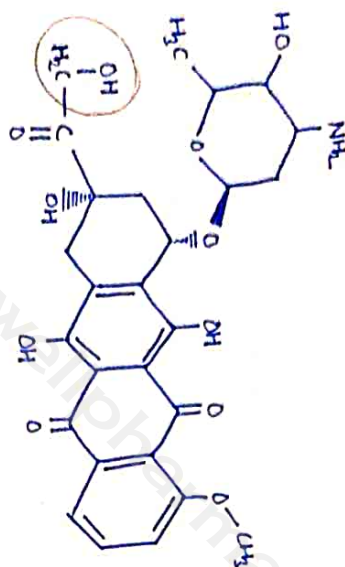
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③

Doxorubicin →

It is a cytotoxic anthracycline antibiotic.
- Obtained from *Streptomyces peucetius*.



• MOA →

- Same as daunorubicin

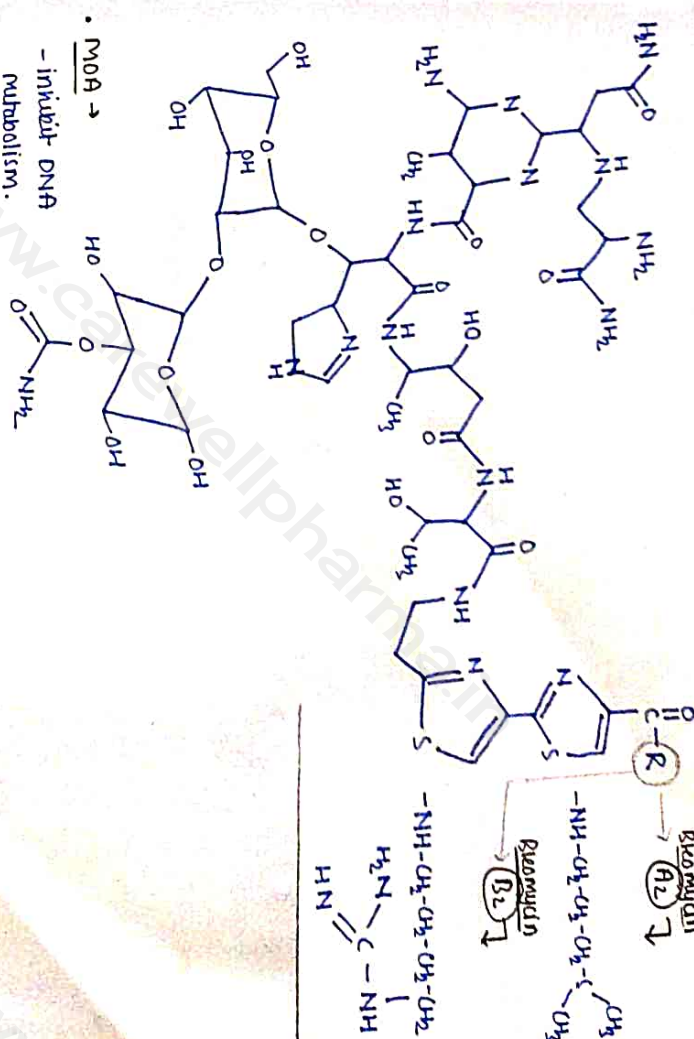
• Uses →

- Used in treatment of leukemia, Kaposi's sarcoma, breast cancer, lymphoma, bladder cancer.

④ Bleomycin →

It is obtained from *Streptomyces verticillus*.

- It consists of bleomycin A₂ and bleomycin B₂.



• MOA →

- inhibit DNA metabolism.

- Same as other.

• Uses →

- Used in the treatment of carcinoma of head, neck, cervix, vulva and penis.

- also used in treatment of trachea, bronchus and lungs..

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④ Plant Products →

Plants medicines are important in the prevention and treatment of cancer.

Because these are derived from plants so they have less undesirable side effects.

— Various drugs →

1) Vinca alkaloids

— isolated from the periwinkle plants

vinca rosea.

- ① Vinblastin
- ② Vincristin

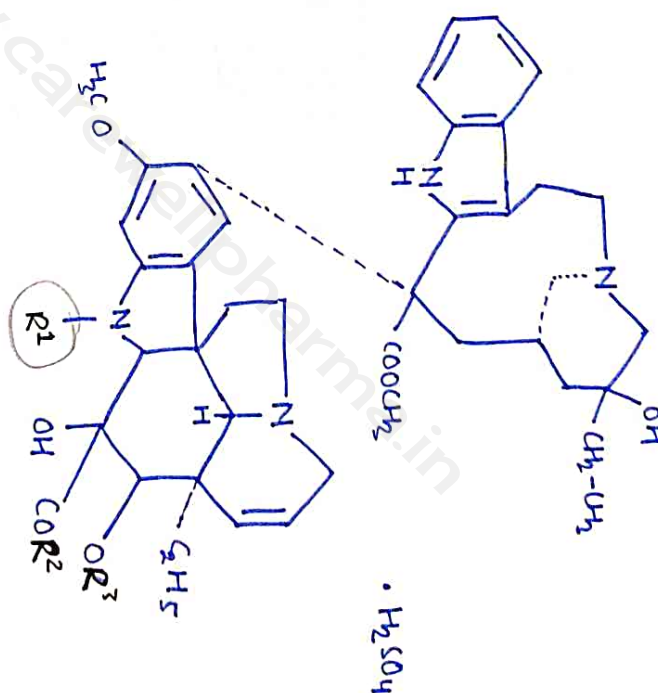
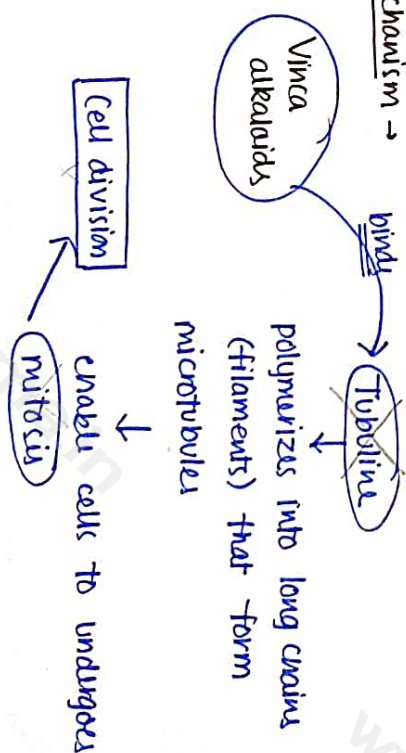
1) Vinblastine sulphate →



2) Vincristine sulphate →



• Mechanism →



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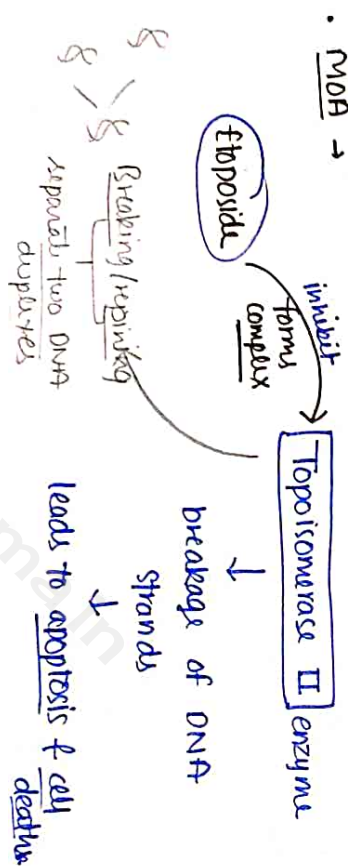
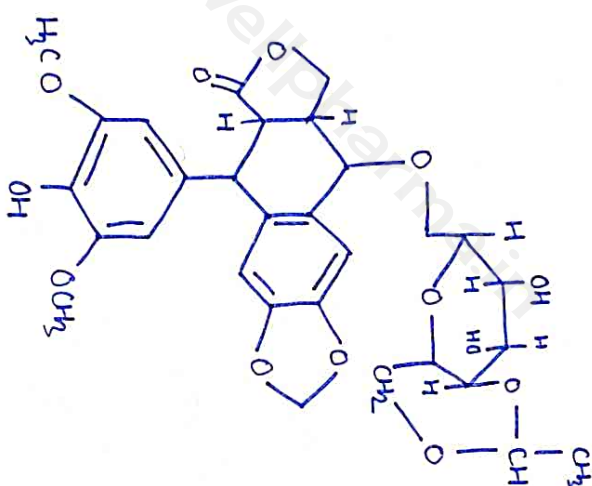
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• Uses →

- used in the treatment of breast cancer, testicular cancer, Hodgkin's lymphomas.
- used in treatment of lymphocytic leukaemia.

3) E-toposide →

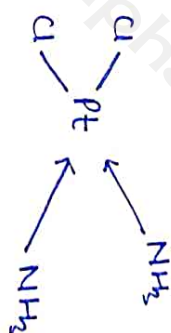


• Uses →

- It is obtained as extracts of many apple plants.
- effective in the treatment of lung cancer, testicular cancer and Hodgkin's disease.
- also used to treat leukemia.

⑤ Miscellaneous

1) Cisplatin →



• MOA →

- same as alkylating agents.

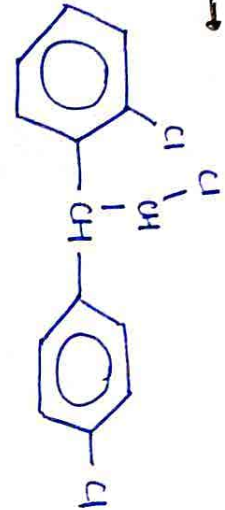
• Uses →

- used in treatment of various cancers include testicular, oesophageal, brain, ovarian, head and neck and breast cancer.
- also used in bladder cancer.

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2) Mitotane →



• MOA →

- still unknown
- But acc to present data, it changes the peripheral metabolism of steroids and directly suppress the adrenal cortex.

• Uses →

- used for treating inoperable adrenocortical tumors and Cushing's syndrome.

